Lab 2: Incident Response Plan Report

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Aim

To design and document a **Cyber Incident Response Plan (CIRP)** for a fictional organization, defining procedures, roles, and controls required to effectively detect, respond to, and recover from cyber security incidents.

Organization Profile (Fictional)

Organization Name: TechNova Solutions Pvt. Ltd.
Industry Domain: IT Services and Software Development
Infrastructure Overview:

- Cloud-based application hosting (AWS/Azure environment)
- Corporate LAN with ~250 endpoints
- Remote workforce connected via VPN
- Web servers, database servers, and DevOps pipelines

Critical Assets:

- Customer data and project repositories
- Source code and intellectual property
- Authentication servers and employee credentials
- Production web applications
- Backup and disaster recovery systems

Security Environment:

- Firewall, IDS/IPS, and endpoint protection deployed
- Role-based access control (RBAC)

- SIEM monitoring and centralized logging
- Regular vulnerability assessments and patch management

Authority and Review: Document Control and Review

Document Control

Field	Details
Author	Cyber Security Analyst Team
Owner	IT Security Department
Date Created	18-02-2026
Last Reviewed By	CISO
Last Date Reviewed	18-02-2026
Endorsed By and Date	CTO – 18-02-2026
Next Review Due Date	18-02-2027

Version Control

Version	Date of Approval	Approved By	Description of Change
1.0	18-02-2026	CTO	Initial CIRP Release

Purpose and Objectives

Purpose of the CIRP

The Cyber Incident Response Plan provides a structured approach to identify, manage, and mitigate cyber incidents to minimize operational disruption, financial loss, and reputational damage.

Objectives of the CIRP

1. Detect and respond to cyber incidents in a timely manner.
2. Protect organizational data and critical infrastructure.
3. Ensure business continuity and rapid recovery.
4. Improve resilience through lessons learned and continuous monitoring.

Standards and Frameworks Considered

This CIRP aligns with globally accepted cyber security standards:

- **National Institute of Standards and Technology (NIST) – NIST SP 800-61 Incident Handling Guide**
- **International Organization for Standardization (ISO/IEC 27001)** for Information Security Management
- **SANS Institute Incident Handler's Handbook**
- **MITRE Corporation ATT&CK Framework** for threat analysis

These frameworks ensure standardized, repeatable, and auditable incident response practices.

High-Level Incident Response Process

1. **Preparation** – Establish tools, policies, and trained response team.
2. **Identification** – Detect and validate incidents using monitoring tools.
3. **Containment** – Limit damage by isolating affected systems.
4. **Eradication** – Remove malware, vulnerabilities, or threat actors.
5. **Recovery** – Restore systems and monitor for recurrence.
6. **Lessons Learned** – Conduct post-incident review and improve controls.

Common Security Incidents and Responses

Common Threats

Threat	Description
External/Removable Media	Malware introduced via USB or external devices.
Attrition	Brute force or DDoS attacks degrading services.
Web	Exploits such as XSS or malicious redirects.
Email	Phishing or malicious attachments.
Supply Chain Interdiction	Compromised hardware/software before deployment.
Impersonation	Spoofing, MITM, rogue access points.
Improper Usage	Policy violations by authorized users.
Loss/Theft	Missing devices containing company data.

Common Incidents

Type / Description	Response
DoS / DDoS	Activate traffic filtering, coordinate with ISP, enable rate limiting.
Phishing	Block sender, reset credentials, conduct awareness alert.
Ransomware	Isolate systems, restore backups, perform forensic analysis.
Malware	Run EDR scan, quarantine files, patch vulnerability.
Data Breach	Disable access, notify management, start legal/compliance process.
ICS Compromise	Disconnect affected systems and initiate forensic containment.

Roles and Responsibilities

Points of Contact for Reporting Cyber Incidents

Name	Hours	Contact	Role Title	Responsibilities
Rajesh Kumar	24/7	security@technova.com	SOC Lead	Initial incident validation
Anita Sharma	Business Hours	itadmin@technova.com	IT Manager	Escalation & coordination

Cyber Incident Response Team (CIRT)

Name	Org Role	Contact	CIRT Role	Responsibilities
Vikram Patel	Security Analyst	vikram@technova.com	Incident Handler	Investigation & containment
Neha Reddy	Network Engineer	neha@technova.com	Network Lead	Traffic isolation
Arjun Das	System Admin	arjun@technova.com	Recovery Lead	Restore services
Priya Nair	Compliance Officer	priya@technova.com	Legal Liaison	Regulatory reporting

Extended CIRT (For Major Incidents)

Name	Org Role	Contact	Role	Responsibilities
CTO	Executive	—	Strategic Lead	Business decisions
HR Head	HR	—	Communication	Employee coordination
PR Manager	Marketing	—	External Comms	Media response

Communications

Internal Communications

- Incident alerts via SIEM dashboard and secure email
- Daily status updates to management
- Internal advisories to employees

External Communications

- Notify customers if services are impacted
- Report to regulatory authorities if required
- Coordinate with law enforcement (if applicable)

Detection, Investigation, Analysis and Activation

Incident Classification

Classification	Description
Critical	Major breach affecting operations or sensitive data.
High	Significant malware or unauthorized access detected.
Medium	Suspicious activity with limited impact.
Low	Policy violations or minor alerts.